



PHYSICS EQUATION SIMULATOR

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JUSTIFICATION FOR SUSTAINABLE DEVELOPMENT GOAL

TITLE	PRIMARY	SECONDARY	TERITARY
GOAL	04	11	17
TARGET	quality education	sustainable communities	partnership for the goals



Who we are



Name of your Venture: **PHYSICS EQUATION SIMULATOR**

Why ?

Explain why do you want to pursue this Business Idea.



Our motive to pursue this business idea is that we are passionate about educating students in a way where all the syllabus thought to us can be applied in real world

What ?

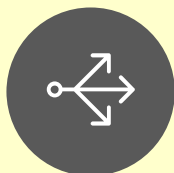
Provide a brief on what does your venture do.



Our venture will ensure that all the complex physics equation can be simulated in real time so that the students can grasp ideas well and deep knowledge of the related topic

How ?

Explain how your venture solves the problem and make its revenue.



Our venture solves the problem by making the students ready to apply all the equations and theories thought to them in real world applications and our venture make its revenue by subscription model or product purchase model



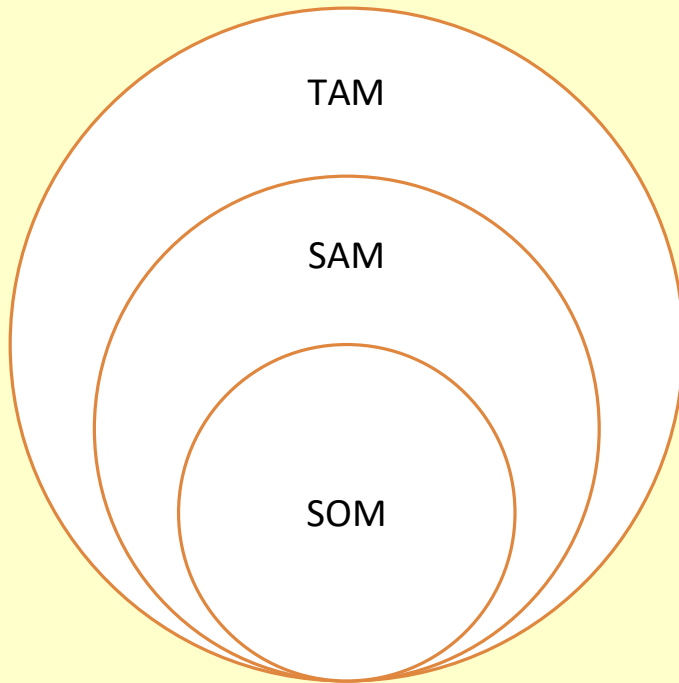
Problem/Opportunity

CONTEXT When does the problem occur? The problem occurs when the student fear the subject and studies syllabus just so they can perform well in their academics rather not focusing on the main objective of the syllabus which is to apply the tough subject in real world applications and solve real world problems	PROBLEM What is the root cause of the problem? Root cause of the problem is that the way that the subject is thought and the intimidation of an subject	ALTERNATIVES What do customers do now to fix the problem? Forced to visualize or rendered videos?
CUSTOMERS Who has the problem most often? students	EMOTIONAL IMPACT How does the customer feel? Better understanding	ALTERNATIVE SHORTCOMINGS What are the disadvantages of the alternatives? run time rendering is not applicable for learning as one would need to apply the concepts in real world problems
	QUANTIFIABLE IMPACT What is the measurable impact (include units)? Decrease in interest in complex concepts 4	

Problem Interviews And Surveys Results

- How many customers did you interview?
- 4
- What was the interview mode?
- one competition and another pitch
- How many of them agree this is a problem and wants a solution?
- 3 agreed this is a problem
- How many of them said they don't need a new solution?
- 0 members

Market Size Estimation



How to calculate market size?

1. Start with Total Addressable market- Global physics students in schools and universities: ~500 million

TAM refers to the total market demand for a product or service.

If you're entering a pre-existing space (like small business banking) you can research it and provide credible sources or reference points on how you arrived at the TAM. If you're creating a new product or space (like Slack), you can estimate the number of customers that would want your product and approximate how much you could charge them.

2. Take your target market (SAM), within that TAM, which varies depending on geography and other logistical factors. Determine the penetration potential of your target market is 950,000 users

3. By conducting research with existing competitors, distributors etc., understand the likely penetration rate in Tamil Nadu could range from 3% to 5%

4. Multiply target market by penetration rate to find your market size if subscription is 1000Rupees/year the low estimate ₹400 million to and high estimate ₹950 million

Sources:



Customer Persona

OUTGOING

GREAT OBSERVER

Approachable



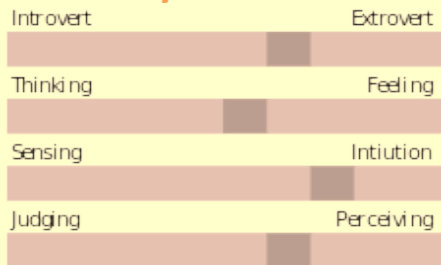
Yogesh

Age: 14-22

Occupation: Student

Location: Coimbatore

Personality



Goals

To become an aeronautical engineer

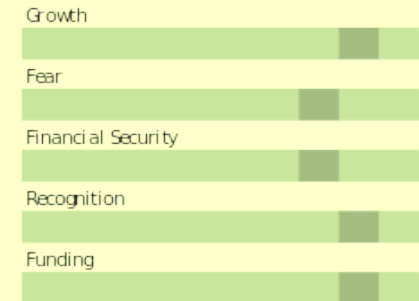
Frustrations

Understands the concepts but doesn't know how to apply them in real world application

Bio

Enthusiastic about upcoming technologies and eager to learn about all the new technologies and their workings, basic application of physics in school projects are his best works

Motivations

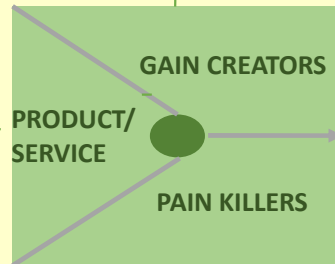


Value Proposition Canvas

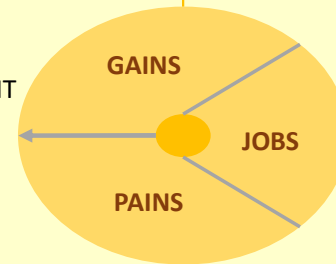
→ What do you offer that makes the customers happy?
→ Genuine interaction between customer that gives them what they need

What is the product or service that you are offering?
Better and easy way to visualize and represent a complex concept which is crucial to real world problems

→ Which features of your offering relieve the customer's pains?
→ real time rendering helps the student to understand the concept and be relieved of fear
→ relieves the customer of fear of intimidating concepts



FIT



I would **LOVE** it if:

→? What would make the customer happy?
→ Their ability to apply the concepts in various projects or personal endeavours
→? What do the clients want when facing the problem?
→ When facing the problem the client would need the assistance of an experienced professional or other source of information

This refers to the feeling/action of customers before he gets in contact with your solution.

I would **WANT**:

→ What do the clients do (actions) when facing the problem?

This refers to the feeling/action of customers before he gets in contact with your solution.

I would **HATE** it if:

→ What are the pains of the clients when facing the problem?

This refers to the feeling/action of customers before he gets in contact with your solution.

Solution

Describe your Solution:

We offer Physics equation simulator

The details of our offering consist of:

1. Visualize of complex concepts
2. Simulations of the complex concepts...
3. Way to help students apply the syllabus in real world problems

List the Benefits of Your solutions

1.Better innovating environment

2.Resource efficient

3.Personalization

Competition Analysis

Benefits	PhET Interactive Simulations	Algodoo	SimScale	Khan Academy	Your Venture
Product	Interactive science simulations	Physics & mechanics simulator	Engineering physics simulations	Educational video tutorials	Real-time physics equation simulator with custom engine
Price	Free	Paid (one-time purchase)	Free with paid options	Free	Subscription model
Branding channels	Educational institutions, website	Education platforms, social media	Engineering communities	Educational institutions, website	Education forums, social media
Packaging	Web-based, easy access	Software application	Cloud-based	Web and app access	Cross-platform app (PC, mobile, web) with Vulkan support
Market reviews	Positive in educational sector	Popular for mechanics learning	Favored for advanced users	Widely used, trusted	Innovative for students seeking hands-on physics learning
UVP	Wide range of science subjects	Visual mechanics exploration	Engineering precision	Free, high-quality content	Real-time, high-performance physics engine with Vulkan
Add more as required	Customization options	Hands-on learning for mechanics	Advanced engineering tools	Interactive learning paths	Game engine-like experience, cross-platform compatibility

Lean Canvas

Problem

Students struggle to understand and apply complex physics equations in real-world contexts. Existing tools are often theoretical and lack interactive, practical application. There is frustration among students who understand concepts but cannot apply them effectively.

EXISTING ALTERNATIVES

Traditional textbooks and online courses that teach theoretical physics concepts. Educational apps and simulation tools, which may not offer customization or real-time applications for complex physics equations.

SOLUTION

A cross-platform physics equation simulator that uses the Vulkan engine to provide high-quality, real-time simulations of complex physics equations.

KEY METRICS

- Subscription and purchase numbers.
- User engagement (e.g., time spent using simulations).
- Student improvement in comprehension and practical skills.

Tailored for students aged 14-22, helping them bridge the gap between theoretical knowledge and real-world application.

UNIQUE VALUE PROPOSITION

"Empowering students to bring physics equations to life, making theoretical concepts practically applicable through real-time, interactive simulations."

HIGH-LEVEL CONCEPT

Bringing physics equations to life for students to enhance real-world understanding and innovation."

UNFAIR ADVANTAGE

Hard-to-replicate knowledge in cross-platform simulation programming, allowing access on multiple devices.

CHANNELS

- Direct sales to students and educational institutions through online educational platforms.

Social media and digital marketing targeting students, educators, and parents.

CUSTOMER SEGMENTS

Students aged 14-22, particularly those in high school and early college focused on physics.

Educational institutions and educators seeking interactive, engaging tools to enhance learning.

EARLY ADOPTERS

Schools or educators interested in adopting advanced, interactive teaching resources.

Business Model

Customer Acquisition
Channels: Social media, partnerships with schools, and influencer marketing.
Target: Students (ages 14-22) and education institutions.
Revenue Streams
Subscription Model: Monthly/annual for students.
One-Time Purchase: Lifetime access option.
Institutional Licensing: Bulk access for schools/universities.
Retention
Personalization: Tailored simulations and regular content updates.
Cross-Platform Access: Usable on any device via Vulkan, ensuring flexibility.
Community: Forum for users to share ideas and support each other.
Scalability
Low incremental costs with global reach.
Potential to expand into other STEM subjects
Key Partnerships
Collaborate with educators, tech providers, and influencers to enhance reach and credibility.

COST STRUCTURE

- Development costs (software licenses, team salaries, platform fees).
- Marketing and promotional expenses.
- Ongoing maintenance and customer support.

REVENUE STREAMS

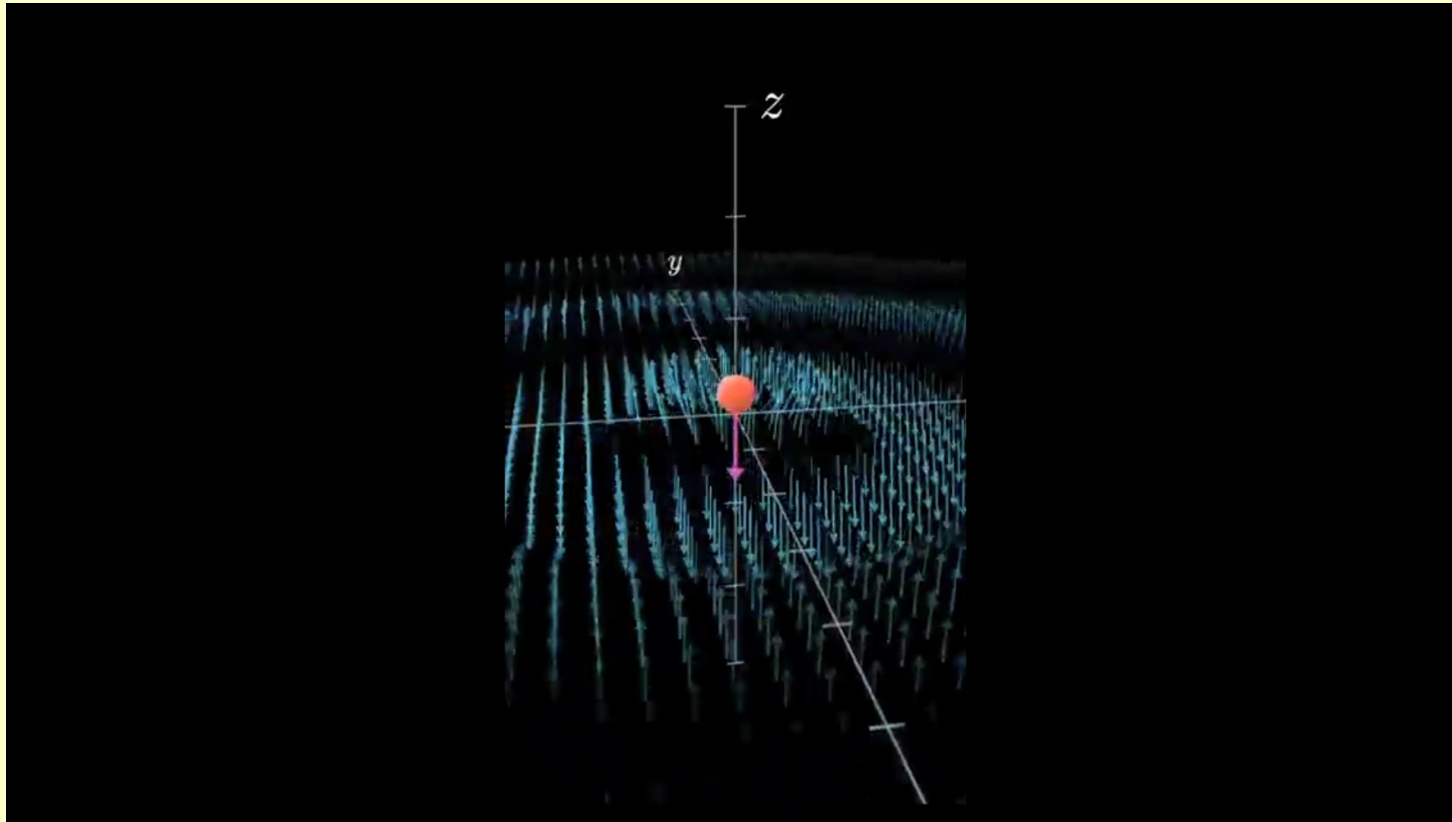
Subscription model (e.g., annual subscription fee for individual users).

- One-time purchase option for full access.

Institutional licensing for schools or universities, allowing multiple students to benefit.

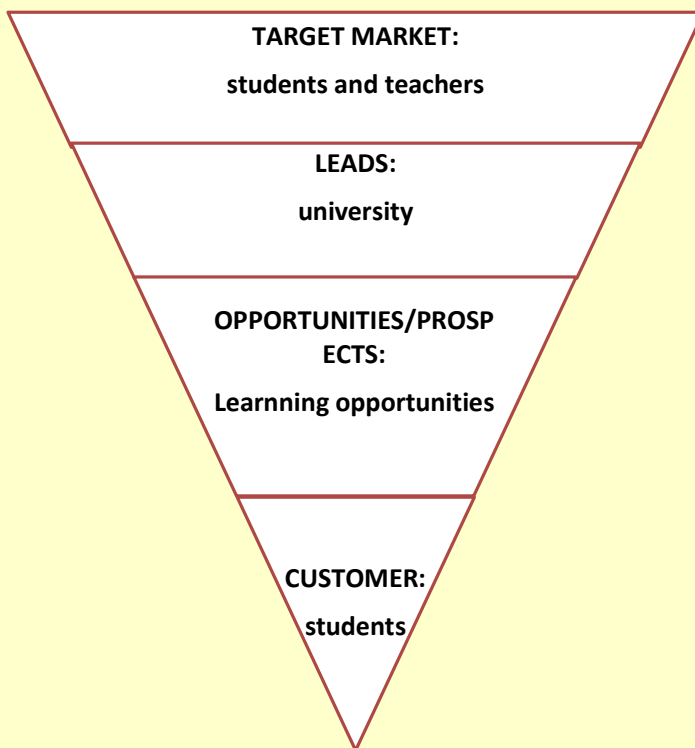


MVP



SALES PLAN

Customer Sales Funnel



Customer Acquisition Plan

1	2	3	4	5
Target Customer Segment (Type)	Target Customer Segment (Number)	Channels to be used to attract the target customer segment	Estimated number of leads	Estimated cost to convert these leads to actual customers
dept HOD	-	showing progress	>100	22,000
An student	73393 43106	product	1	0
university	-	complete product	>2000	57,000

Go-to-Market Strategy

- *Social Media Presence: Establish accounts on Facebook, LinkedIn, Instagram, and Twitter. Regularly post educational content, tutorials, and user testimonials.*
- *Branding Video: Create a 1-2 minute video that highlights the product, showcasing its unique value of making physics learning more practical and interactive. The video should include:*
 - *What the simulator is*
 - *The target audience (students and educators)*
 - *Key benefits: interactive, real-time simulations, and easy-to-use interface*
 - *Positioning Statement: "Physics Equation Simulator is designed for students and educators to simplify and visualize complex physics concepts, helping bridge the gap between theory and practical application."*

Start-up Costs

Starting Costs	
Startup Cost	₹17,48,000
Business Registration	₹25,000
Licenses	₹23,000
Equipment and Utesils	₹6,00,000
Interirors and Furnitures	₹6,00,000
Website development	₹5,00,000
Capital Work in Progress (Fixed Asset)	
Laptop/ POS	
Starting Operations (Budgeted)	₹4,52,000
Promotion	₹1,00,000
Hosting and Domain Space	₹2,00,000
Misceleneous	₹1,00,000
Start-up capital	₹22,00,000
Share Captial	₹12,00,000
	₹10,00,000



Forecast P&L



COGS													
Cost of Services	₹ 3,00,000	₹ 3,00,000	₹ 3,00,000	₹ 3,00,000	₹ 3,00,000	₹ 3,00,000	₹ 3,00,000	₹ 3,00,000	₹ 3,00,000	₹ 3,00,000	₹ 3,00,000	₹ 3,00,000	₹ 36,00,000
Cost Per Unit	₹ 100	₹ 100	₹ 100	₹ 100	₹ 100	₹ 100	₹ 100	₹ 100	₹ 100	₹ 100	₹ 100	₹ 100	₹ 100
Gross profit	₹ 12,00,000	₹ 12,00,000	₹ 12,00,000	₹ 12,00,000	₹ 12,00,000	₹ 12,00,000	₹ 12,00,000	₹ 12,00,000	₹ 12,00,000	₹ 12,00,000	₹ 12,00,000	₹ 12,00,000	₹ 1,44,00,000
Fixed Expenses													
Salaries	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 91,20,000
Marketing and Promotion													₹ -
Digital Marketing													₹ -
Utilities (Electricity etc.)													₹ -
Customer Service													₹ -
Rent													₹ -
Total Expenses	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 91,20,000
Result													
Net Profit/Loss	₹ 4,40,000	₹ 4,40,000	₹ 4,40,000	₹ 4,40,000	₹ 4,40,000	₹ 4,40,000	₹ 4,40,000	₹ 4,40,000	₹ 4,40,000	₹ 4,40,000	₹ 4,40,000	₹ 4,40,000	₹ 52,80,000
Gross Profit Margin	80%	80%	80%	80%	80%	80%	80%	80%	80%	80%	80%	80%	80%
Net Profit Margin	29%	29%	29%	29%	29%	29%	29%	29%	29%	29%	29%	29%	29%



Financial Projections

	Year 1	Year 2	Year 3	Year 4	Year 5
Number of sales	36000	43206	51847	62221	74625
Value of each sale (Contribution)	₹ 400	₹ 400	₹ 486	₹ 486	₹ 536
Average Price per unit	₹ 500	₹ 500	₹ 600	₹ 600	₹ 650
Average Cost per unit	₹ 100	₹ 100	₹ 114	₹ 114	₹ 114
Total revenue	₹ 1,80,00,000	₹ 2,16,03,000	₹ 3,11,08,320	₹ 3,73,32,384	₹ 4,85,04,607
Gross Profit	₹ 1,44,00,000	₹ 1,72,82,400	₹ 2,51,82,704	₹ 3,02,21,187	₹ 3,99,75,742
Capital costs					
Land and building	N.A.	N.A.	N.A.	N.A.	N.A.
Equipment					
Product development costs					
Others					
Total					
Expenses (Annual)					
Salaries	₹ 91,20,000	₹ 1,00,32,000	₹ 1,10,35,200	₹ 1,21,38,720	₹ 1,33,52,592
Marketing and Promotion	₹ -	₹ -	₹ -	₹ -	₹ -
Digital Marketing & Customer Services	₹ -	₹ -	₹ -	₹ -	₹ -
Utilities (electricity etc.)	₹ -	₹ -	₹ -	₹ -	₹ -
Office Supplies	₹ -	₹ -	₹ -	₹ -	₹ -
Rent	₹ -	₹ -	₹ -	₹ -	₹ -
Total	₹ 91,20,000	₹ 1,00,32,000	₹ 1,10,35,200	₹ 1,21,38,720	₹ 1,33,52,592
Earnings (EBITDA)	₹ 52,80,000	₹ 72,50,400	₹ 1,41,47,504	₹ 1,80,82,467	₹ 2,66,23,150

UNIT ECONOMICS

P & L/ unit	Year 2024		Unit Economics	Year 2024			Step 1	APV = Average Purchase Value = Total revenue/Total number of purchases	₹ 500		
Revenue	₹ 500		CAC	₹ -			Step 2	APF = Average Purchase Frequency = Number of purchases/ Number of unique customers	144.00		
COGS	₹ 100		CLV	₹ 11,520			Step 3	Average Customer Lifespan = 1/Churnrate	0.20		
Gross Profit	₹ 400		ARPU	₹ 72,000			Step 4	CLV =APV*APF*ACL*GM	₹ 11,520		
Gross Profit Margin	80%										
Operating Costs	₹ 353										
Operating Profit	₹ 146.67										
			CAC = (M + S) / Number of Customers								
			1000000	666.6666667							
Assumptions			1500								
Churn rate	customers at the beginning - customers at the end/ customers at the beginning	5									
Customers at the Beginning of month	1000										
Customers at the End of month	950										

Funding Plan

How much funds required to reach to the next level of the venture?

- 3,00,000

How much have been bootstrapped? If not, why ?

- 1,000, not much since the state the start up is at early research phase

How much can be bootstrapped?

- 12,000

How much external funding required? If not, why?

- 12,00,000

Funds utilization strategy (Details)

- Software & Licenses
- Cloud Computing, GPU Resources
- Bug Fixes & Optimization
- Building an Intuitive UI
- Support for Different Devices
- Documentation & Tutorials
- Social Media & Outreach
- Partnerships & Collaborations – Universities, research institutions, or ed-tech companies.
- Customer Support
- One-Time License Fee
- Ads & Sponsorships

Thank You!